

Solution Architecture

Date	Team ID	Project Name
30 September 2026	SWTID-2026-9981	PocketSmart AI: Your Smart Budget & Recommendation Assistant

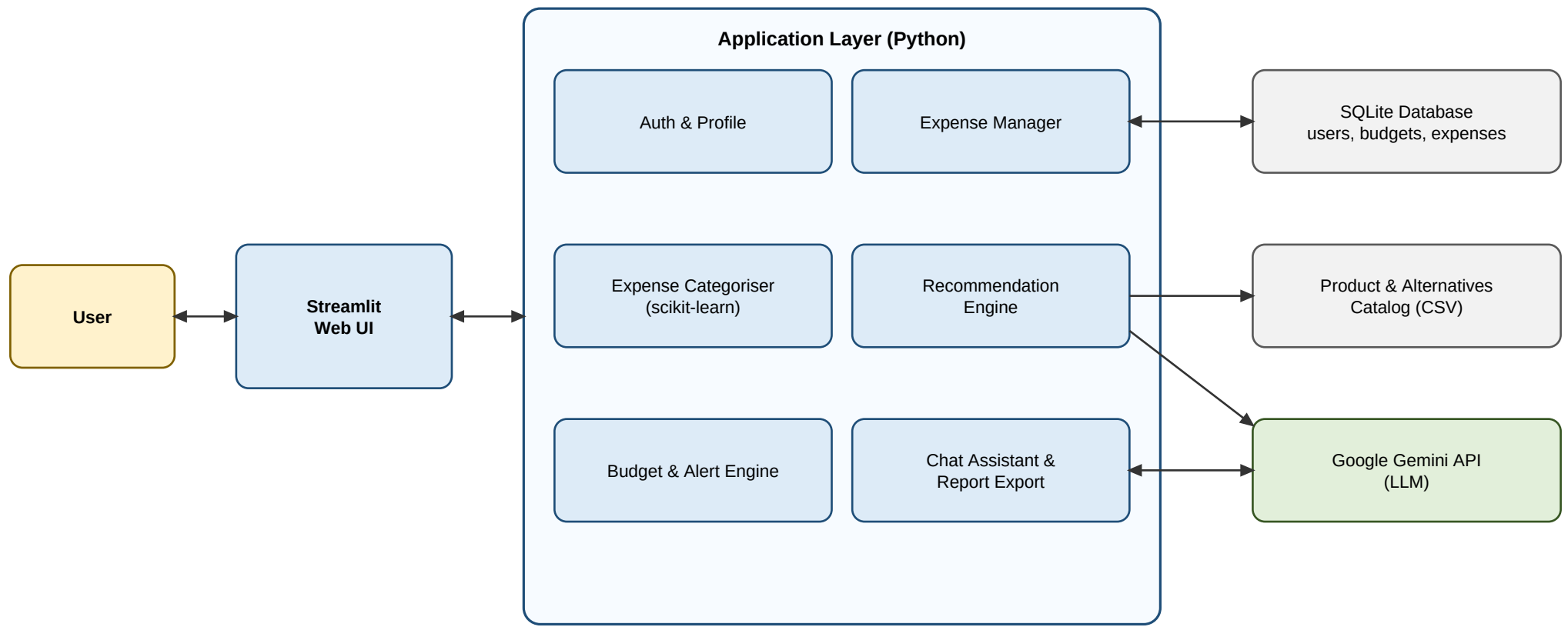


Figure: Solution architecture of PocketSmart AI

Component Description

Layer	Component	Responsibility
Presentation	Streamlit Web UI	Collects user input; shows dashboard, budgets, recommendations, chat and reports.
Application	Auth & Profile	Registration, login, income and savings goal.

Layer	Component	Responsibility
Application	Expense Manager	Add, edit, delete and import expenses.
Application	Expense Categoriser	TF-IDF and Logistic Regression model predicts the category of each expense.
Application	Budget & Alert Engine	Compares spending with category limits and raises 80 percent and 100 percent alerts.
Application	Recommendation Engine	Builds a spending summary, adds catalogue items within the remaining budget and prompts the LLM for tips.
Application	Chat Assistant & Report Export	Answers questions from the user's data; exports the monthly CSV.
Data	SQLite Database	Persistent storage for users, budgets and expenses.
Data	Product & Alternatives Catalog	Curated list of items, prices and cheaper alternatives used for suggestions.
External	Gemini API	Generates natural-language advice and chat answers.

Request Flow

Step	Action
1	User adds an expense through the web UI.
2	Expense Categoriser predicts the category; the record is saved in SQLite.
3	Budget & Alert Engine recalculates spending against limits and shows any alert.
4	On request, Recommendation Engine sends a spending summary and remaining budget to the LLM together with matching catalogue items.
5	The response is shown to the user as tips and budget-friendly suggestions.